3D Model

Step-by-step documentation of the modeling process

Jonathan Felipe López Nuñez
jolopezn@unal.edu.co

Professor: Andres Ricardo Arevalo Murillo

November 6, 2026

Contents

1 Introduction **2**

2 Reference setup **2**

3 Body **3**
3.1 Base shape 3
3.2 Head 4

4 Ears **5**

5 Feet and claws **5**

6 Arms and hands **6**

7 Belly **7**

8 Materials **7**

9 Headband **8**

10 Eyes **9**

11 Mouth and fangs **10**

12 Headband plate symbol **11**

13 Rivets **12**

14 Summary of tools and modifiers **13**

1 Introduction

This document presents the step-by-step 3D modeling workflow of Snorlax in Blender, developed for the creation of the final `.obj` model and its multi-material setup.

The visual concept and modeling were inspired by the character design featured in the video *"[Blender Character Modeling] Snorlax Joined the Akatsuki"* by Poly Crafted (Figure 1). The original project scope intended to include both the Akatsuki cloak and the Magikarp sword. However, because a significant amount of time was invested in learning Blender fundamentals, tools, modifiers, and modeling techniques, the work was concentrated on achieving a clean, detailed model of Snorlax complete with the rogue ninja headband, metal plate, and engraved symbol.



Figure 1: Inspiration model from the YouTube video *"[Blender Character Modeling] Snorlax Joined the Akatsuki"* by Poly Crafted.

2 Reference setup

Before modeling, reference drawings of the character were loaded into the scene so that every part could be traced with the correct proportions.

1. Add → Image → Reference was used to import a front view and a side view of the character.
2. The front image was aligned with the *front orthographic* view (`Numpad 1`) and the side image with the *right orthographic* view (`Numpad 3`), both centered on the world origin.
3. The references were grouped in a collection called *Referencia* so they could be hidden or locked easily, and their opacity was lowered so the model is visible through them.

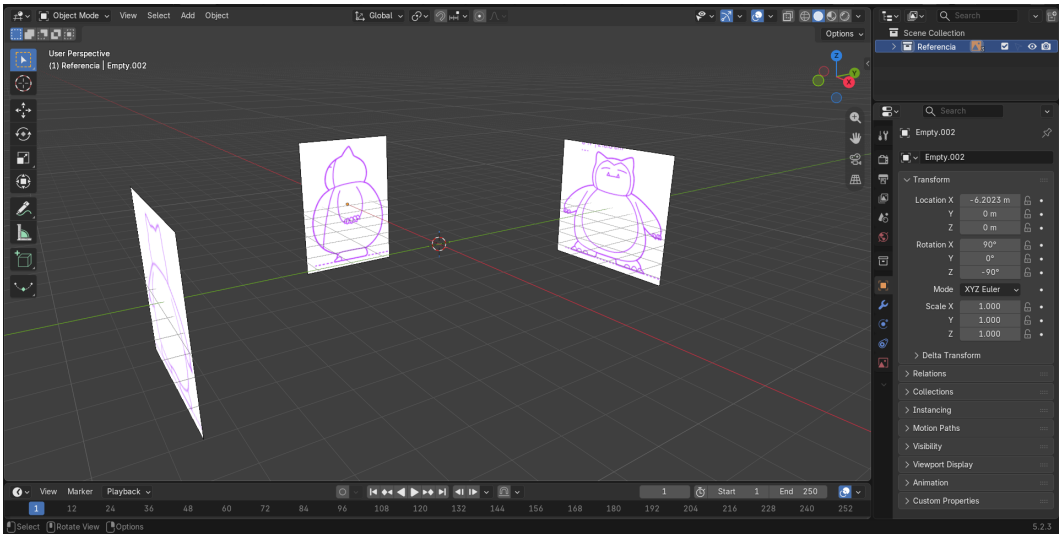


Figure 2: Reference images imported into the scene.

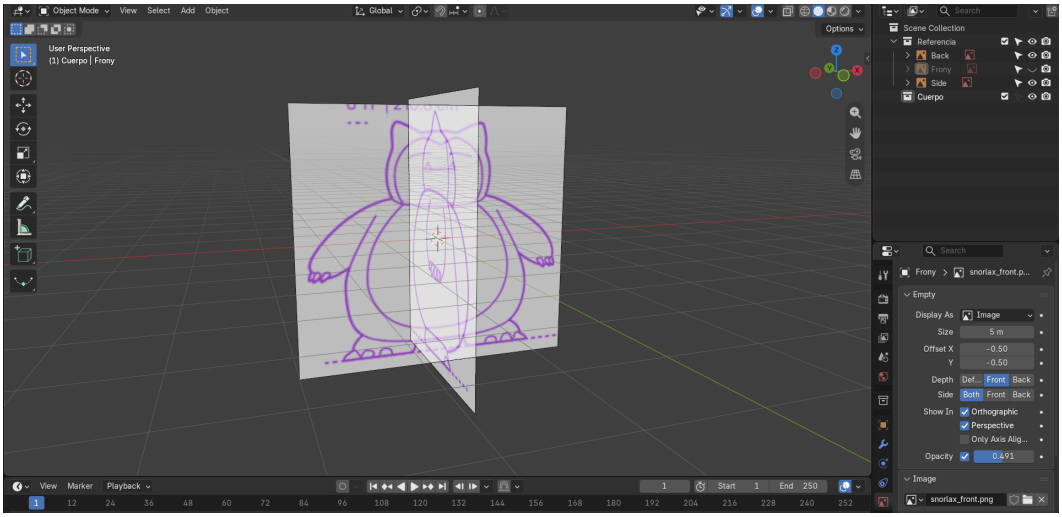


Figure 3: Front and side references aligned at the origin.

3 Body

3.1 Base shape

1. A cube was added (Add → Mesh → Cube) and scaled to fit the outline of the body in the front view.
2. A **Subdivision Surface** modifier was added so the cube becomes a smooth, rounded volume while keeping a low number of editable vertices.
3. In **Sculpt Mode**, the *Grab* brush was used with symmetry on the X axis to push and pull the surface until the silhouette matched the reference: a wide, pear-shaped body that is heavier at the bottom.



Figure 4: Initial cube placed over the front reference.

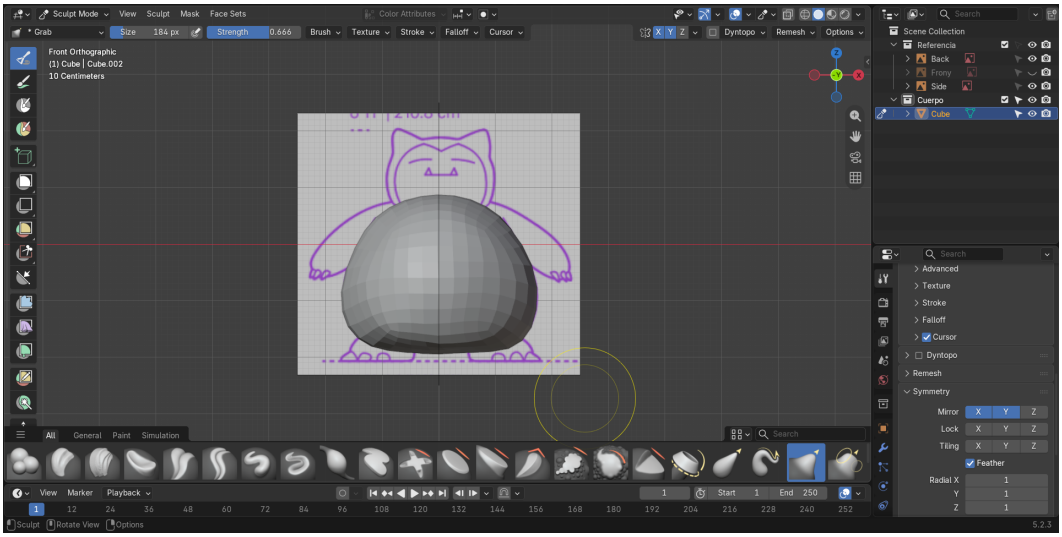
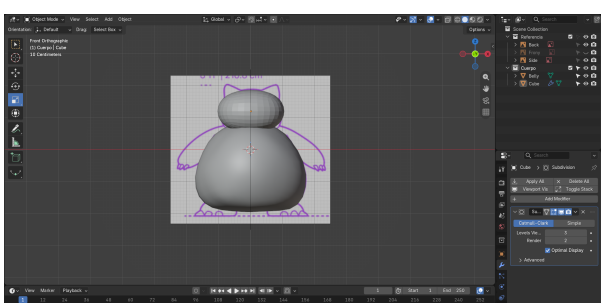


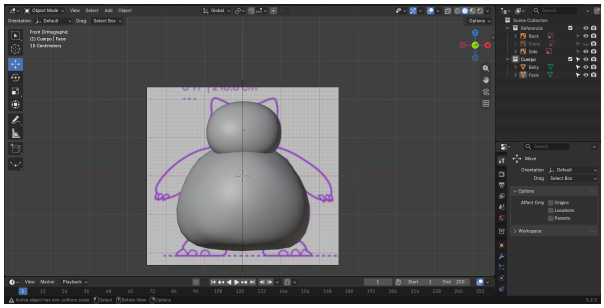
Figure 5: Shaping the body with the Grab brush in Sculpt Mode (X symmetry enabled).

3.2 Head

A second rounded object was added above the body and adjusted in the same way (scaling in Object Mode and editing vertices in Edit Mode) until it matched the head outline in both views.



(a) Head volume added.

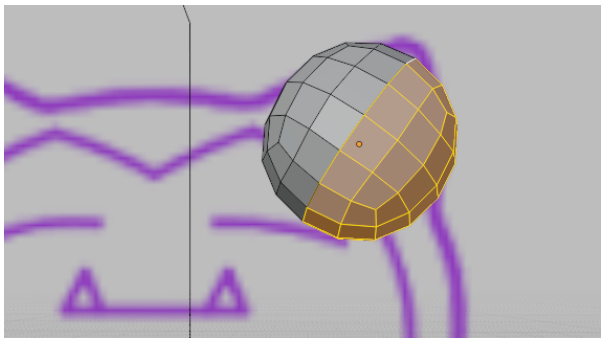


(b) Body and head blocked out.

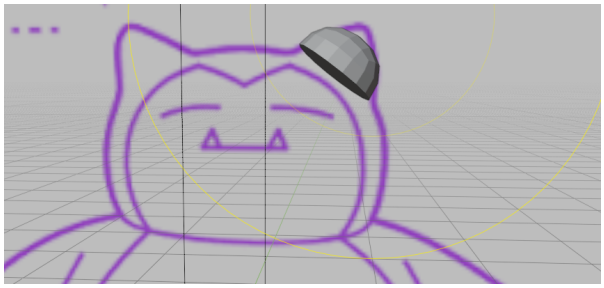
Figure 6: Blocking out the main volumes.

4 Ears

1. A UV sphere was added near the top of the head.
2. In Edit Mode, half of the sphere was selected and removed so that only a shell remained; this shell is the base of the ear. (An attempt to use *Rip* (`V`) failed with the error *"Cannot rip non-manifold vertices or edges"*, so the faces were deleted instead.)
3. The shell was scaled and rotated to form the pointed ear shape and placed on the head.
4. A **Mirror** modifier on the X axis created the second ear automatically.



(a) Half of the sphere selected.

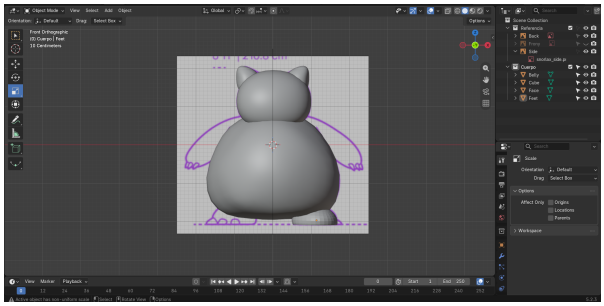


(b) Ear placed on the head.

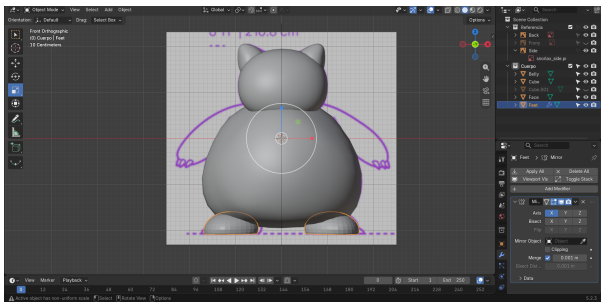
Figure 7: Modeling the ears.

5 Feet and claws

1. Each foot was made from a flattened sphere placed at the bottom of the body, following the side reference.
2. A **Mirror** modifier (X axis) duplicated the foot on the other side so both feet always stay symmetric.
3. The toes were modeled with small cubes. In Edit Mode, the tip faces were selected and extruded with `E` (*Extrude Region and Move*) to create pointed claws.

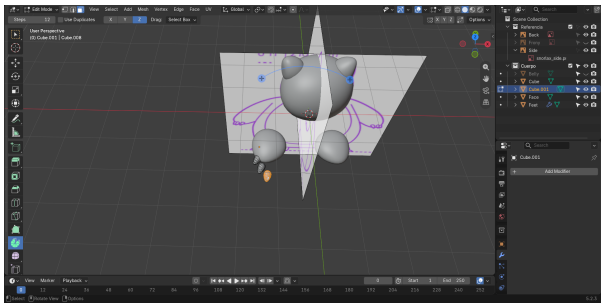


(a) Feet added.

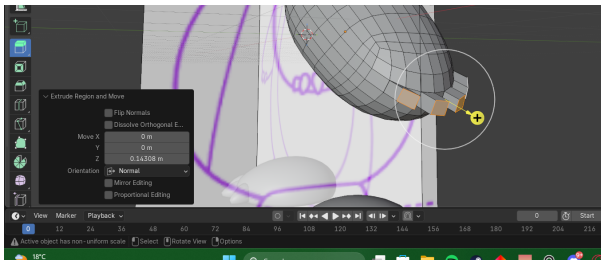


(b) Mirror modifier for symmetry.

Figure 8: Feet.



(a) Toes positioned with symmetry.

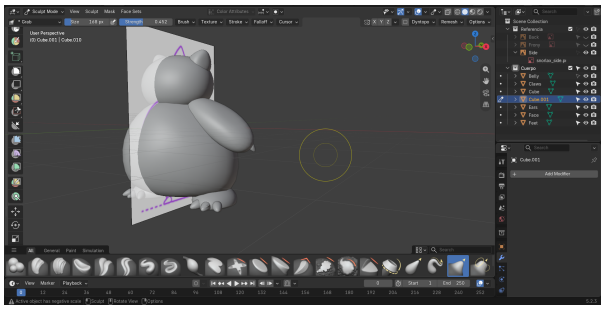


(b) Extruding the claw tips (E).

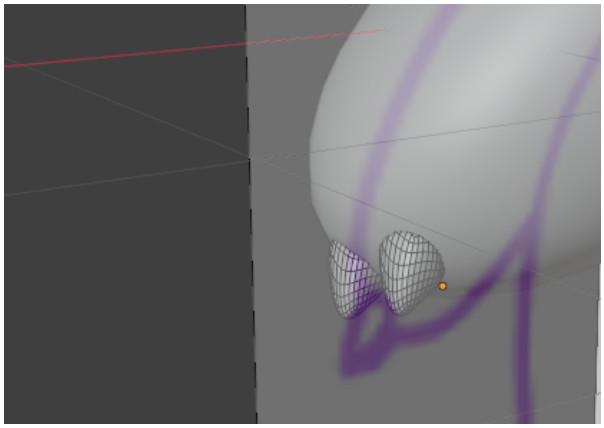
Figure 9: Toes and claws.

6 Arms and hands

1. Each arm started as an elongated sphere rotated to follow the reference.
2. In Sculpt Mode the arm was shaped with the Grab brush to make it thicker at the shoulder and thinner at the wrist.
3. Small claws were added at the end of each hand using the same extrusion technique as the feet.
4. The arm was mirrored to the other side.



(a) Sculpting the arm.



(b) Hand claws.

Figure 10: Arms.

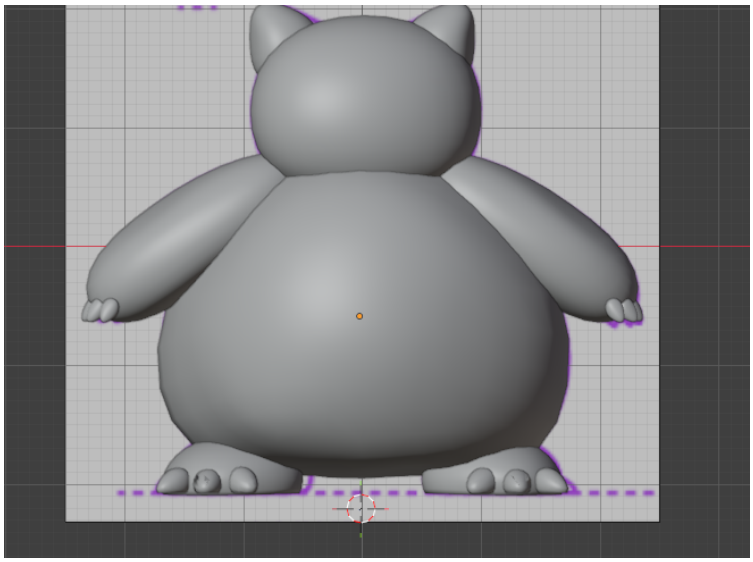
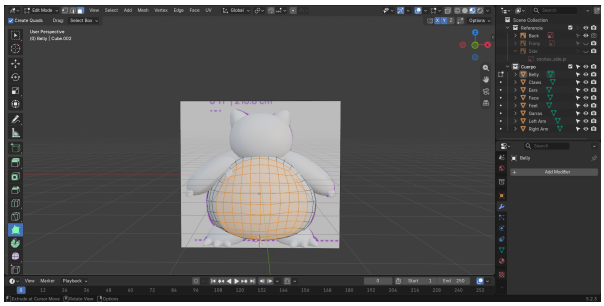


Figure 11: Complete body with arms, feet and ears.

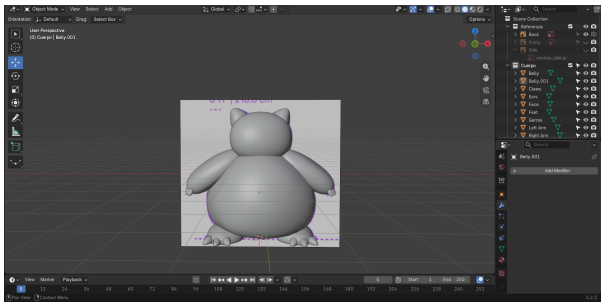
7 Belly

The belly needs a different color from the rest of the body.

1. In Edit Mode with face selection (`3`), the front faces of the body that form the belly were selected.
2. They were duplicated and separated into their own object (`Shift+D`, then `P` → Selection), so the belly can receive its own material and be adjusted independently.



(a) Belly faces selected.



(b) Belly as a separate object.

Figure 12: Belly.

8 Materials

Materials were created in the Material Properties tab using the *Principled BSDF* shader:

- **Body** – blue, used for head, body, arms and ears.
- **Belly / face** – off-white, used for the belly, face, feet and claws.
- **Metal** – grey with high *Metallic* and medium *Roughness*, used for the headband plate and rivets.
- **Black** – used for the eyes, mouth and plate symbol.

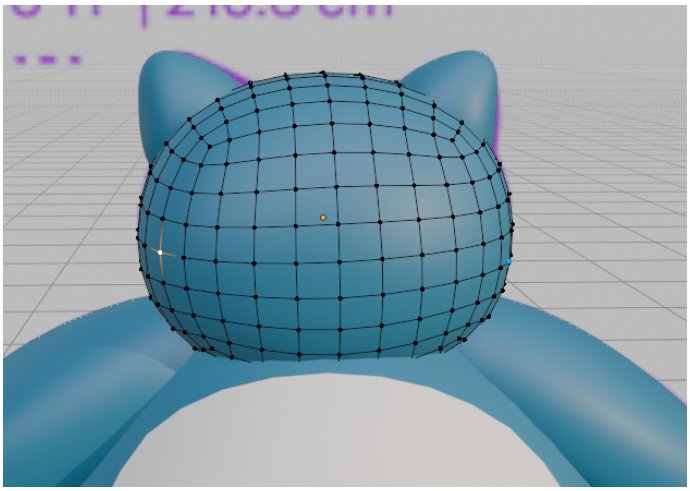
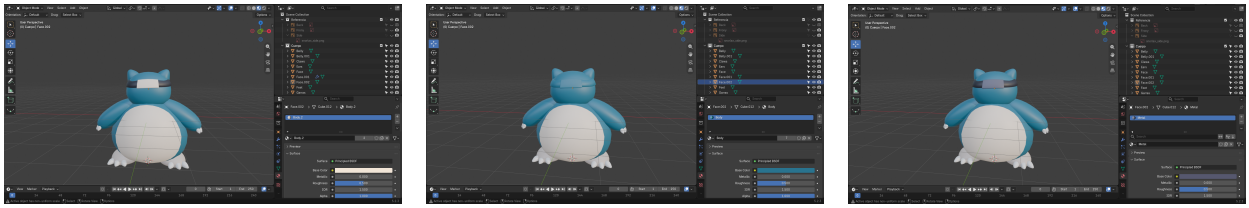


Figure 13: Head with its material and visible topology.



(a) Off-white material. (b) Blue material. (c) Metal material.

Figure 14: Materials applied to the model.

9 Headband

1. In Edit Mode, a horizontal ring of faces around the forehead of the head mesh was selected.
2. These faces were duplicated and separated into a new object (`Shift+D`, `P` → Selection). Because the band was created from the head itself, it follows the curvature of the head perfectly.
3. A **Solidify** modifier gave the band thickness, and a **Bevel** modifier rounded its edges.
4. A separate block was modeled in front for the metal plate and given the *Metal* material.

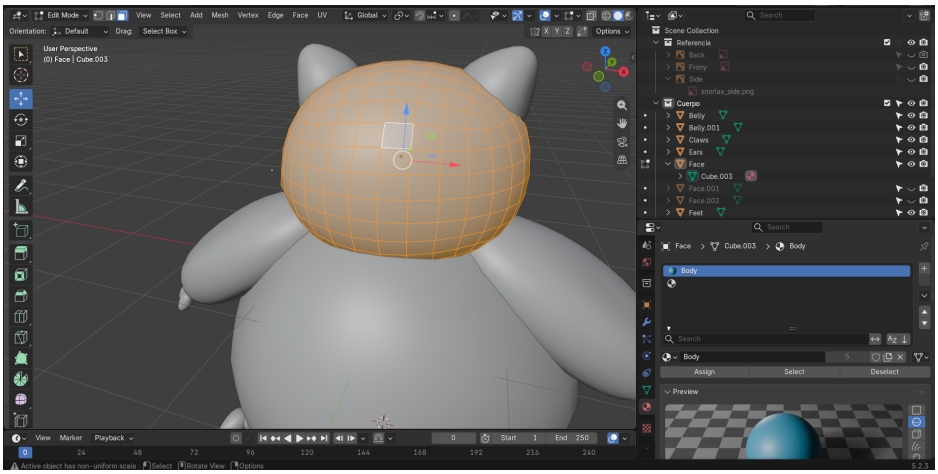
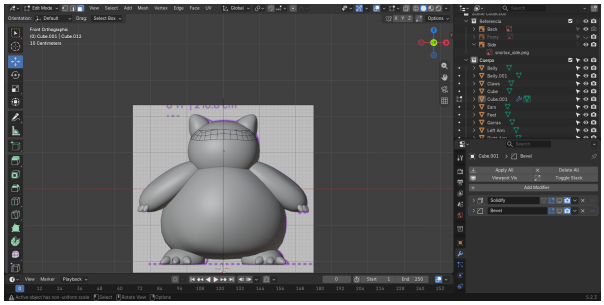
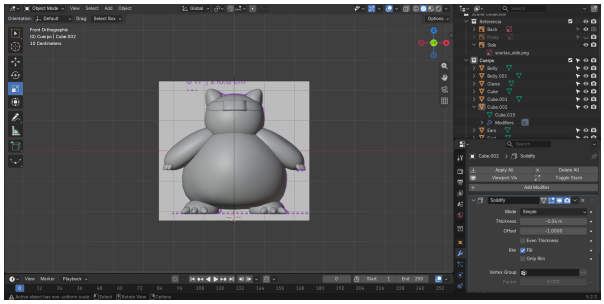


Figure 15: Faces of the head selected to create the headband.



(a) Solidify and Bevel modifiers.



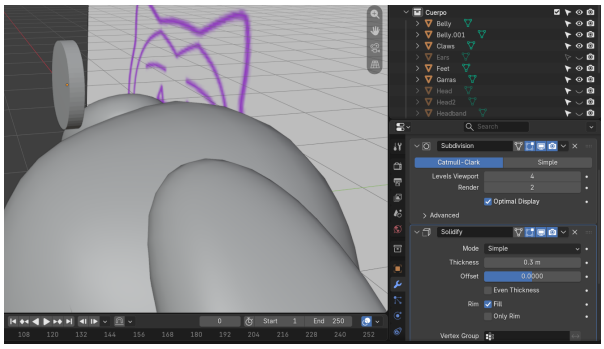
(b) Solidify settings.

Figure 16: Headband.

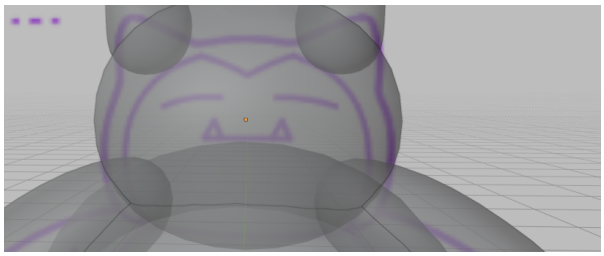
10 Eyes

The eyes are thin curved strips that must sit exactly on the curved surface of the face.

1. A plane was added, rotated to face the front and shaped in Edit Mode into a thin strip following the eye line of the reference.
2. The plane was subdivided (right click → Subdivide) so it has enough vertices to bend. A plane with only four vertices cannot follow a curved surface.
3. Scale and rotation were applied (`Ctrl+A` → All Transforms) so the modifiers calculate thickness and direction correctly.
4. A **Shrinkwrap** modifier was added with the head as *Target*, *Wrap Method: Project* along the Y axis (positive and negative) and an *Offset* of 0.004 m so the strip rests on the skin without sinking in.
5. A **Solidify** modifier (thickness 0.03 m, offset +1) gave it volume towards the outside of the face.
6. The modifier order was set to *Subdivision* → *Shrinkwrap* → *Solidify*: the strip is first smoothed, then wrapped onto the face, and only then thickened.

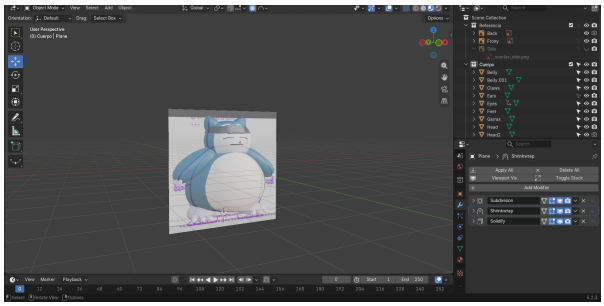


(a) Eye plane with Subdivision and Solidify.

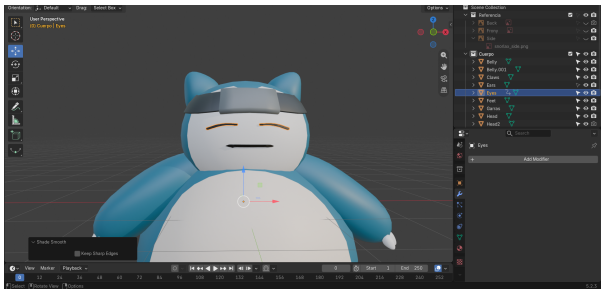


(b) X-Ray view to align with the reference.

Figure 17: Preparing the eyes.



(a) Shrinkwrap modifier added.

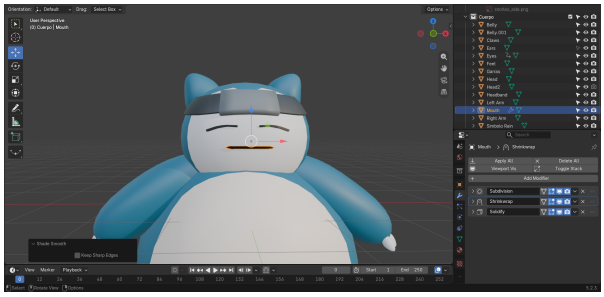


(b) Eyes resting on the face.

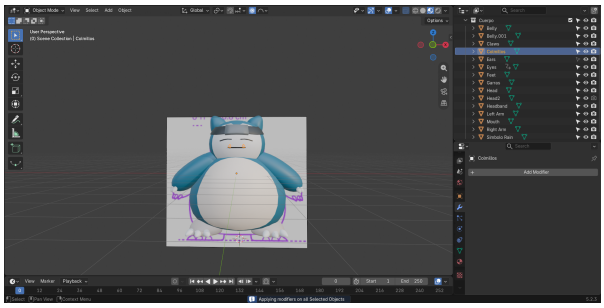
Figure 18: Eyes.

11 Mouth and fangs

1. The mouth was made the same way as the eyes: a thin subdivided strip with Subdivision, Shrinkwrap (target: head) and Solidify.
2. The two fangs were made from cones with *Radius 2* set to 0 so they end in a point. They were scaled, placed at the edge of the mouth and rotated so they point outward, following the angle of the face.
3. Shade Auto Smooth was applied to all the face pieces.



(a) Mouth with Shrinkwrap.



(b) Fangs added.

Figure 19: Mouth and fangs.

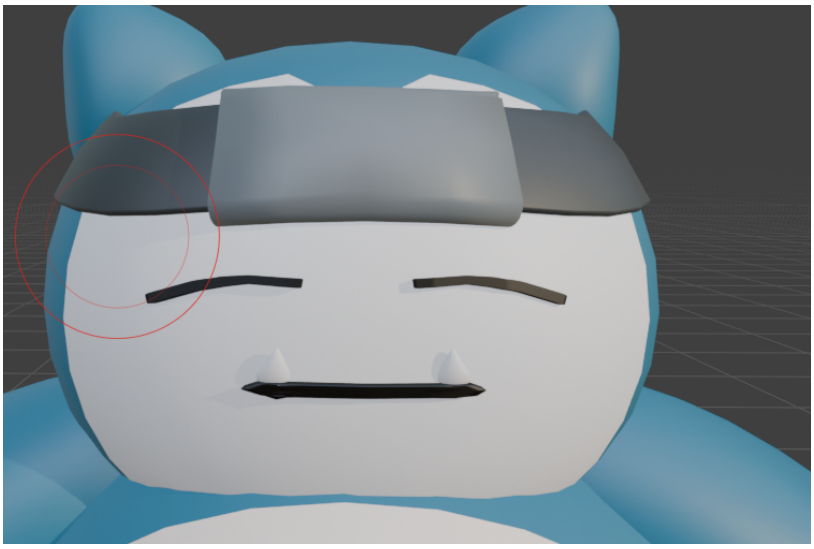
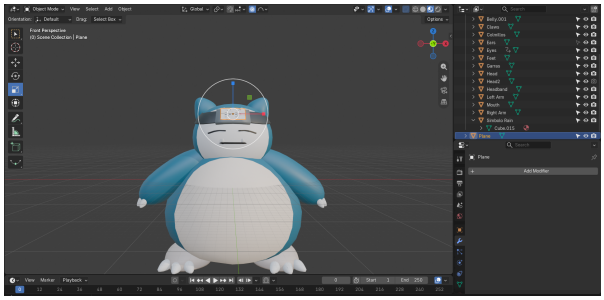


Figure 20: Finished face: eyes, mouth and fangs.

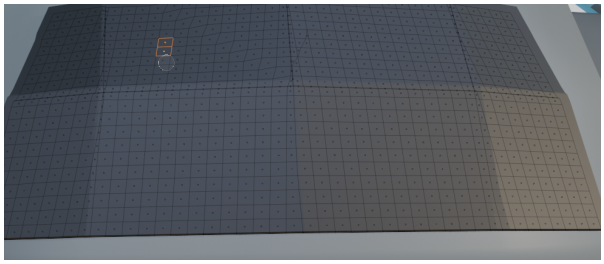
12 Headband plate symbol

The symbol on the plate was made with a "pixel" technique: a grid sheet that is already wrapped onto the plate, from which the unneeded squares are deleted.

1. A grid plane (*Lamina_ Dibujo*) of 42×22 squares was placed in front of the plate.
2. It received a **Shrinkwrap** modifier with the plate as target (*Nearest Surface Point*, *Outside Surface*, offset 0.003 m) and a **Solidify** modifier (0.012 m) so it follows the curvature of the plate and has relief.
3. In Edit Mode, front view (`Numpad 1`) and face selection (`3`), the squares that form the symbol were selected: the circle-select tool (`C`) for free shapes and `Ctrl+click` (shortest path) for straight lines.
4. The selection was inverted (`Ctrl+I`) and the remaining squares were deleted (`X` → `Faces`). Only the symbol remains, already curved and with thickness.

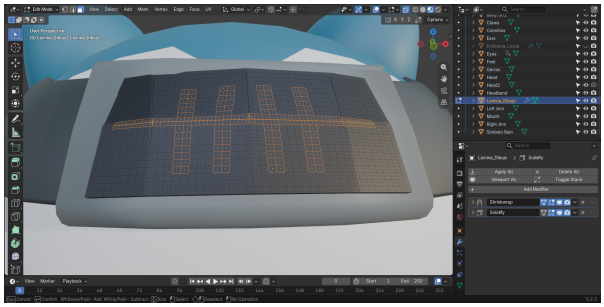


(a) Grid sheet on the plate.

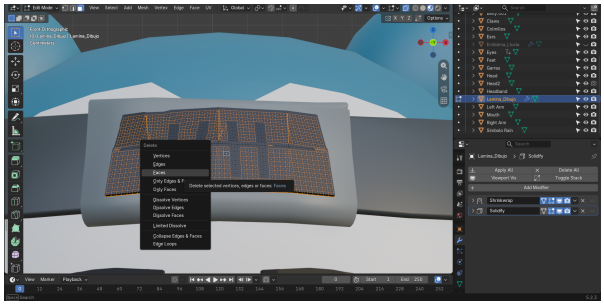


(b) Close-up of the grid.

Figure 21: Preparing the symbol sheet.



(a) Symbol squares selected.



(b) Deleting the inverted selection.

Figure 22: Drawing the symbol.

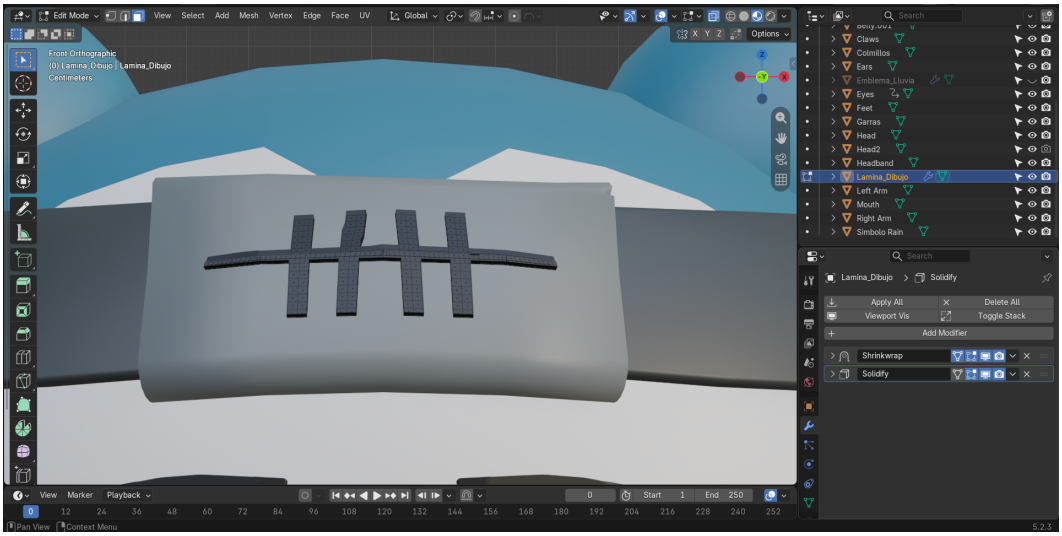


Figure 23: Finished symbol on the plate (Edit Mode, front view).

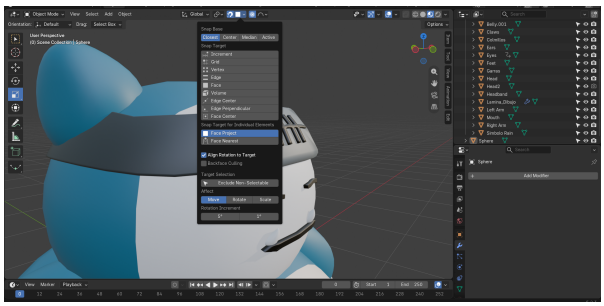
13 Rivets

Rivets were added at both ends of the plate as separate objects, so the plate mesh did not need to be modified.

1. A UV sphere (16 segments, 8 rings, small radius) was added and flattened along its depth axis so it looks like a rivet head.
2. Shade Auto Smooth and the *Metal* material were applied, and the transforms were applied (`Ctrl+A`).
3. Snapping was enabled (`Shift+Tab`) with *Snap To: Face Project* and *Align Rotation to Target*, so that moving the rivet with `G` makes it stick to the plate and orient itself to the surface.
4. The rivet was duplicated (`Shift+D`) to form a vertical column of three, and the copies were joined into one object (`Ctrl+J`).
5. A **Mirror** modifier with the plate as *Mirror Object* created the column on the opposite end symmetrically.

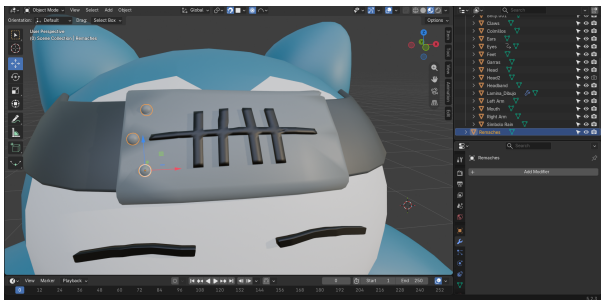


(a) UV sphere for the rivet.

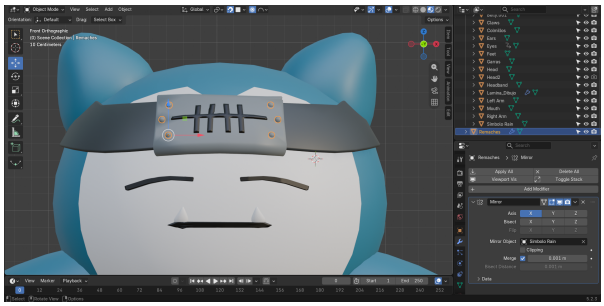


(b) Snapping settings.

Figure 24: Creating and placing a rivet.



(a) Column of rivets.



(b) Mirror modifier: final result.

Figure 25: Rivets.

14 Summary of tools and modifiers

Tool / modifier	**Used for**
Reference images	Tracing proportions in front and side views
Sculpt Mode (Grab)	Body, head and arm shapes
Extrude (`E`)	Claws on feet and hands
Separate (`P`)	Belly and headband from existing meshes
Mirror	Ears, feet, arms and rivets
Subdivision Surface	Smooth, rounded surfaces
Solidify	Thickness of headband, eyes, mouth and symbol
Bevel	Rounded edges of the headband
Shrinkwrap	Eyes, mouth and symbol following curved surfaces
Snapping (Face Project)	Placing rivets on the plate
Shade Auto Smooth	Smooth shading of the final pieces

Table 1: Main tools used in the project.